

Practice Quiz: Iterative Refinement

Part A: Multiple Choice

1. The action-perception loop in prototyping consists of: A) Designing, manufacturing, selling, iterating B) Building a prototype (action), testing it (perception), updating beliefs, and selecting the next change (action) C) Planning, building, launching, marketing D) Hypothesizing, funding, building, patenting
2. The minimum viable prototype is defined as: A) The cheapest possible prototype B) The simplest prototype that generates meaningful prediction errors for the highest-priority hypothesis C) The first prototype an inventor builds D) A prototype with no working features
3. A/B testing is valuable because it: A) Doubles the cost of prototyping B) Provides controlled comparison between design alternatives, isolating the effect of a single variable C) Eliminates the need for user feedback D) Always produces a clear winner
4. The pivot-or-persevere decision is fundamentally about: A) Whether to increase or decrease the budget B) Whether to continue parameter refinement within the current model or switch to a structurally different model C) Whether to hire more engineers D) Whether to file a patent
5. Diminishing returns in iteration (each iteration producing smaller improvements) is a signal to: A) Work harder on the same approach B) Consider whether a structural model change (pivot) is needed rather than further parameter tuning C) Abandon the project entirely D) Increase the prototype's fidelity
6. Iteration roadmaps prevent ad hoc tinkering by: A) Eliminating creativity B) Specifying hypothesis priority, decision gates, and resource allocation for each iteration C) Locking in the final design before testing D) Removing the need for testing

7. SpaceX's Starship development philosophy of "build, fly, fail, learn, repeat" prioritizes:
- A) Hardware preservation
 - B) High loop frequency and information gain from each test, even at the cost of hardware loss
 - C) Avoiding all risk
 - D) Simulating before building

Part B: Short Answer

1. Explain why Drew Houston's 3-minute Dropbox video was a more effective minimum viable prototype than building the actual file synchronization technology. Use the concepts of Markov blanket scope, hypothesis priority, and information-to-cost ratio in your answer.
2. You have built three iterations of a new bicycle pedal design. v1 was a basic shape test (cardboard), v2 added cleat compatibility (3D print), and v3 added material testing (aluminum). Each iteration resolved a specific uncertainty. Design iterations v4 and v5, specifying the hypothesis, fidelity level, and decision gate for each.
3. A team has been iterating on a social media app for 12 months. Downloads have plateaued at 2,000. The team is divided: half want to pivot to a different app concept, half want to persevere with improved features. Using the expected free energy framework, describe how you would analyze this decision. What evidence would you need to see to recommend one path over the other?